

Synthesis Review: Rates, Extrema, and the Shape of a Graph

Answer Key

AP Calculus AB/BC

Quick Answers

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|--|---|
| 1. 62.83 | 6. (a) V in terms of $h = \frac{\pi h^3}{12}$, (b) $dh/dt = 0.16$ |
| 2. (a) $A'(x) = 24 - 2x$, (b) $x = 12$ | 7. (a) critical x -values = 1, 4, — |
| 3. (a) $f'(x) = 3x^2 - 12x + 9$, (b) critical x -values = 1, 3, (c) inflection x -value = 2 | 8. 100 |
| 4. (a) $S(x) = x^2 + \frac{128000}{x}$, (b) base edge $x = 40$ | 9. (a) $A'(x) = 24 - 6x^2$, (b) critical values = -2, 2, (c) maximum area = 32 |
| 5. (a) $f''(x) = 12x^2 - 12$, (b) inflection x -values = -1, 1, — | 10. (a) $dr/dt = 0.08$, (b) $dS/dt = 20$, — |
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What is verified

19 of 22 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 5, 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

CHA-3.D – problems 1, 6, 8, 10

FUN-4.A – problems 3, 5, 7

FUN-4.B – problems 2, 4, 9

Difficulty 1–5 of 5 across 22 tagged checks.

Common wrong answers

8. If they got 140: added the two speeds instead of differentiating the distance relation

1. Oil slick, $\frac{dr}{dt} = 0.5$ m/s. Find $\frac{dA}{dt}$ when $r = 20$ m.

Solution. The relation is $A = \pi r^2$; differentiate with respect to time, then substitute.

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(20)(0.5) = 20\pi$$

$$\frac{dA}{dt} \approx 62.83$$

2. Garden, 48 m of fence, $A(x) = x(24 - x)$. (a) $A'(x)$. (b) The maximizing x .

Solution (a). Expand first: $A(x) = 24x - x^2$.

$$A'(x) = 24 - 2x$$

Solution (b). $24 - 2x = 0$.

$$x = 12$$

Then the other side is $24 - 12 = 12$ as well: the maximizing rectangle is a square, and its area is 144 square metres.

3. $f(x) = x^3 - 6x^2 + 9x + 2$. (a) $f'(x)$. (b) Critical x -values. (c) Inflection x -value.

Solution (a).

$$f'(x) = 3x^2 - 12x + 9$$

Solution (b). $3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0$.

$$x = 1 \text{ or } x = 3$$

Solution (c). $f''(x) = 6x - 12 = 0$, and f'' changes sign there.

$$x = 2$$

4. Open-top box, square base x , volume 32000 cm³, $S = x^2 + 4xh$. (a) $S(x)$. (b) The minimizing x .

Solution (a). The constraint $x^2h = 32000$ gives $h = \frac{32000}{x^2}$, so $S = x^2 + 4x \cdot \frac{32000}{x^2}$.

$$S(x) = x^2 + \frac{128000}{x}$$

Solution (b). $S'(x) = 2x - \frac{128000}{x^2} = 0$ gives $2x^3 = 128000$, so $x^3 = 64000$.

$$x = 40$$

Since $S''(x) = 2 + \frac{256000}{x^3} > 0$ for $x > 0$, this is a minimum.

5. $f(x) = x^4 - 6x^2 + 5$. (a) $f''(x)$. (b) Inflection x -values. (c) Classify $x = 0$.

Solution (a). $f'(x) = 4x^3 - 12x$, so

$$f''(x) = 12x^2 - 12$$

Solution (b). $12x^2 - 12 = 0$, and the concavity changes at each root.

$$x = -1 \text{ or } x = 1$$

Solution (c) — instructor-judged. $f'(0) = 0$, so the origin is a critical point, and $f''(0) = -12 < 0$, so the curve is concave down there: $x = 0$ gives a local maximum. A sign chart for f' (positive to the left of 0, negative to the right) earns equal credit. A response that stops at “ $f'(0) = 0$ ” has not classified anything.

6. Inverted cone, $\frac{dV}{dt} = 2 \text{ m}^3/\text{min}$, $r = \frac{h}{2}$. (a) V in terms of h . (b) $\frac{dh}{dt}$ when $h = 4 \text{ m}$.

Solution (a). Substituting $r = \frac{h}{2}$ into $V = \frac{1}{3}\pi r^2 h$ removes r before any differentiation:

$$V = \frac{\pi h^3}{12}$$

Solution (b). Differentiating, $\frac{dV}{dt} = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{4}{\pi h^2} \cdot \frac{dV}{dt}$. At $h = 4$ this is $\frac{4(2)}{\pi(4)^2} = \frac{1}{2\pi}$.

$$\frac{dh}{dt} \approx 0.16$$

7. $g'(x) = (x-1)^2(x-4)$. (a) Critical x -values. (b) What happens to g at each.

Solution (a). g' is zero where either factor vanishes.

$$x = 1 \text{ or } x = 4$$

Solution (b) — instructor-judged. The factor $(x-1)^2$ is never negative, so the sign of g' is the sign of $(x-4)$: negative for $x < 4$ (except at $x = 1$, where it is zero) and positive for $x > 4$. So g' does *not* change sign at $x = 1$ — the graph flattens and keeps falling, no extremum — while at $x = 4$ it changes from negative to positive, a local minimum. Full credit needs both conclusions with the sign reasoning; a graph sketch alone does not.

8. Two cars, north at 60 km/h and east at 80 km/h; after 2 hours they are 120 km and 160 km out, 200 km apart. Find the rate the distance changes.

Solution. With x east and y north, $x^2 + y^2 = D^2$, so $x\frac{dx}{dt} + y\frac{dy}{dt} = D\frac{dD}{dt}$ and

$$\frac{dD}{dt} = \frac{(160)(80) + (120)(60)}{200} = \frac{12800 + 7200}{200}.$$

$$\frac{dD}{dt} = 100$$

Kilometres per hour. Note that this is not $80 + 60$: the distance grows more slowly than the sum of the speeds because the cars move at right angles.

9. Rectangle under $y = 12 - x^2$, $A(x) = 24x - 2x^3$. (a) $A'(x)$. (b) Solutions of $A'(x) = 0$. (c) Maximum area.

Solution (a).

$$A'(x) = 24 - 6x^2$$

Solution (b). $24 - 6x^2 = 0$ gives $x^2 = 4$.

$$x = -2 \text{ or } x = 2$$

Solution (c). Only $x = 2$ is in the region $x > 0$; there the rectangle is 4 wide and 8 tall, so $A(2) = 48 - 16$.

$$A(2) = 32$$

10. Balloon, $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$. (a) $\frac{dr}{dt}$ at $r = 10$. (b) $\frac{dS}{dt}$ there. (c) Compare with $r = 5$.

Solution (a). $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$, so $\frac{dr}{dt} = \frac{100}{4\pi(10)^2} = \frac{1}{4\pi}$.

$$\frac{dr}{dt} \approx 0.08$$

Solution (b). $\frac{dS}{dt} = 8\pi r \frac{dr}{dt} = 8\pi(10) \left(\frac{1}{4\pi} \right)$.

$$\frac{dS}{dt} = 20$$

Square centimetres per second — and the π cancels, which is a good sign the setup is right.

Solution (c) — instructor-judged. Combining the two steps, $\frac{dS}{dt} = 8\pi r \cdot \frac{1}{4\pi r^2} \frac{dV}{dt} = \frac{2}{r} \frac{dV}{dt}$, which *decreases* as r grows. At $r = 5$ the rate is 40, twice the value at $r = 10$, so the surface area grows faster at the smaller radius. Full credit needs the comparison and a reason; naming the larger number alone does not.